



IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom



UE E – AML

Online competition : Stock Return Prediction by QRT

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Groupe 13

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Competition Context

Objective: predict the **sign** of next-day residual stock return

Residual return = stock return **without** market effect

Inputs:

- Past 20 days of residual returns
- Past 20 days of relative volumes
- Stock metadata (sector, industry)

Metric:

→ **Accuracy** on return direction
(up / down)

Project Structure & Pipeline

- **Reproducible architecture**
- **Separation of concerns:** data, notebooks, reusable code, reports
- **Notebooks for exploration** → **scripts for production ready logic**
- **Clear pipeline:** EDA → Baseline_model → F_I → Advanced models
- **Easier collaboration and debugging**

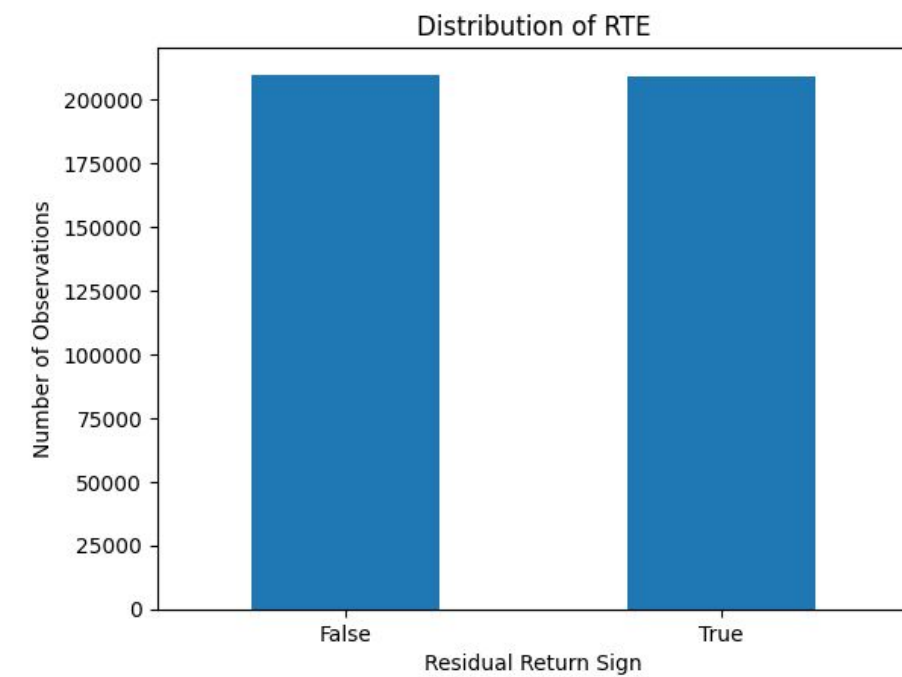
```
market-neutral-alpha/  
├── data/  
│   ├── raw/           # Original challenge CSVs  
│   └── processed/      # Cleaned & feature-engineered data  
├── notebooks/  
│   ├── 01_eda_financial.ipynb  
│   ├── 02_baseline_model.ipynb  
│   ├── 03_feature_engineering.ipynb  
│   ├── 04_models.ipynb  
│   └── 05_evaluation-metrics.ipynb  
├── src/  
│   ├── features/  
│   │   └── build_features.py  
│   ├── models/  
│   │   └── train_model.py  
│   ├── evaluation/  
│   │   └── evaluate.py  
├── submissions/  
│   └── submission_v1.csv  
├── pyproject.toml  
├── README.md  
├── .gitignore  
└── .env
```

II- Exploratory DATA Analysis 1/2

◆ Target & Raw Signals

i) Target distribution:

- Binary target: next day residual return sign
- Classes are **perfectly balanced**

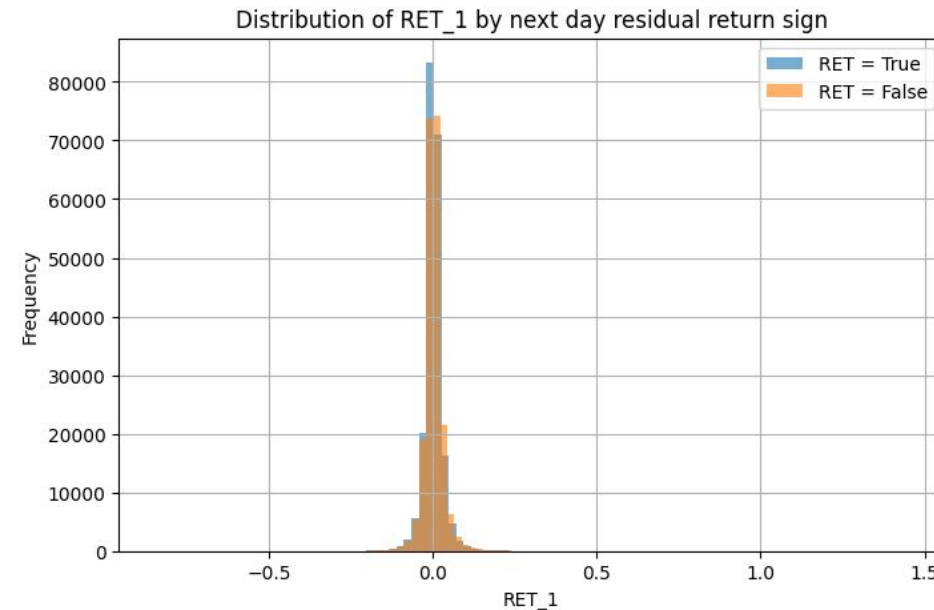


ii) Raw return characteristics

Daily residual returns:

- **centered around zero**
- **heavy-tailed**
- **extremely noisy**

CC : Directional prediction cannot rely on raw returns alone.



II- Exploratory DATA Analysis 2/2

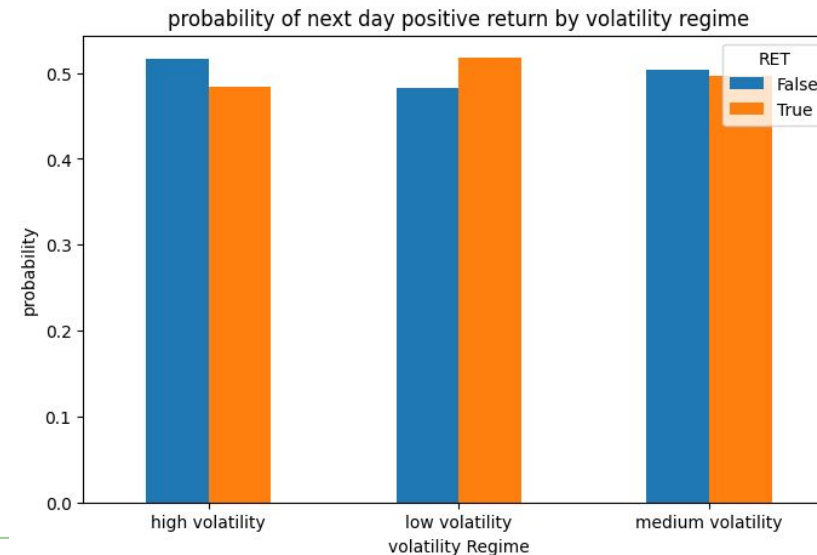
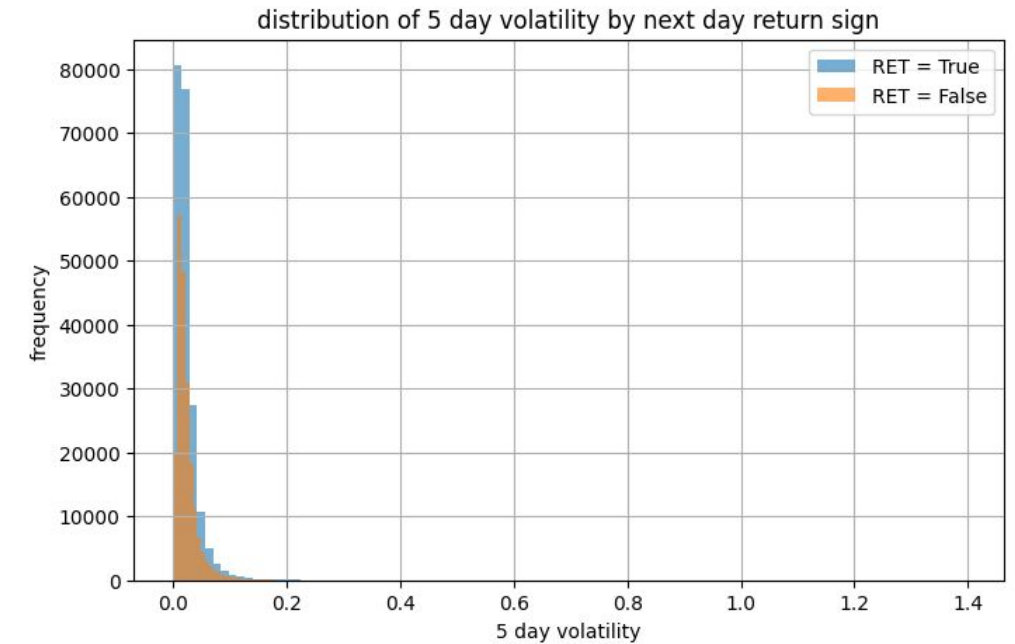
❖ Volatility Structure

I) VOLATILITY IS :

- **strictly positive**
- **right-skewed**
- **heteroskedastic (non constant VAR of a dependent variable)**

→ **low / medium / high volatility regimes**

Next day return probability varies across volatility regimes

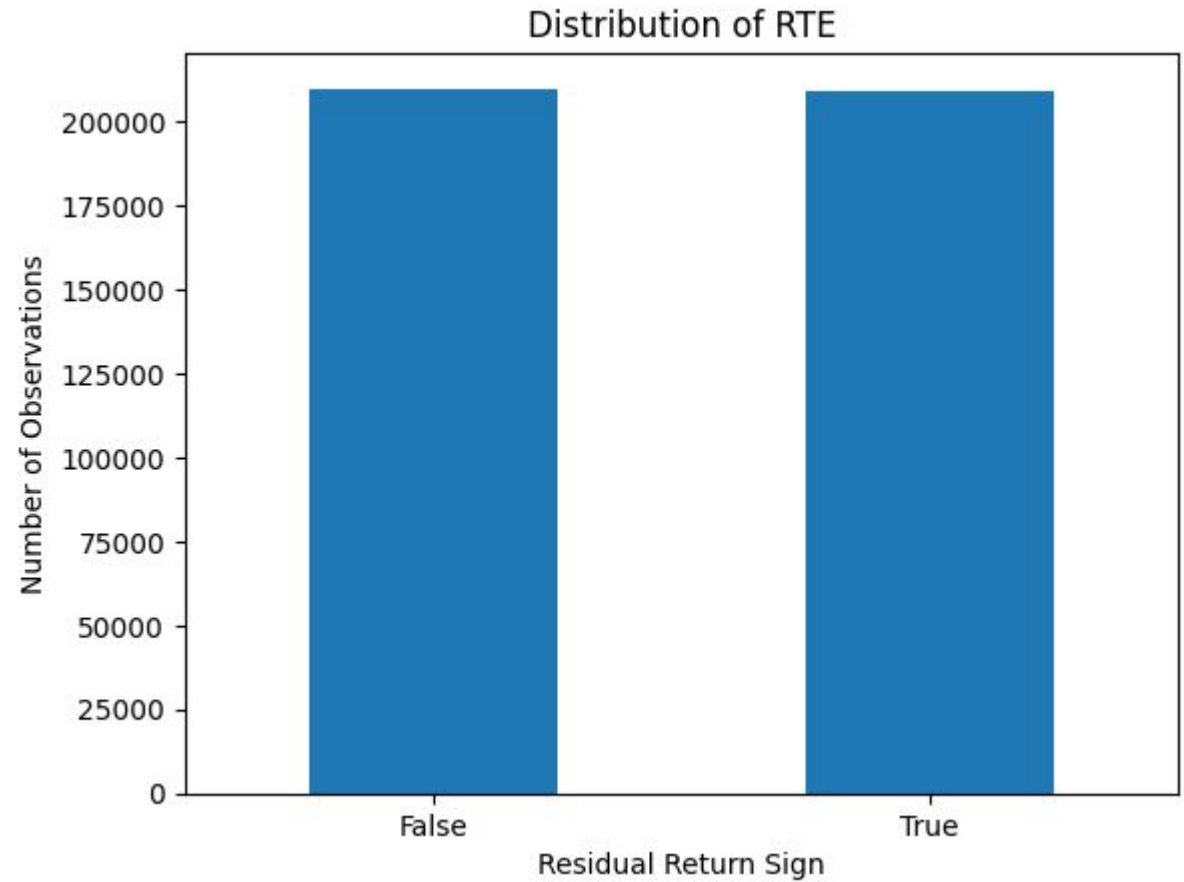


II - Baseline

❖ Basic Logistic Regression

ACCURACY : 50.5%

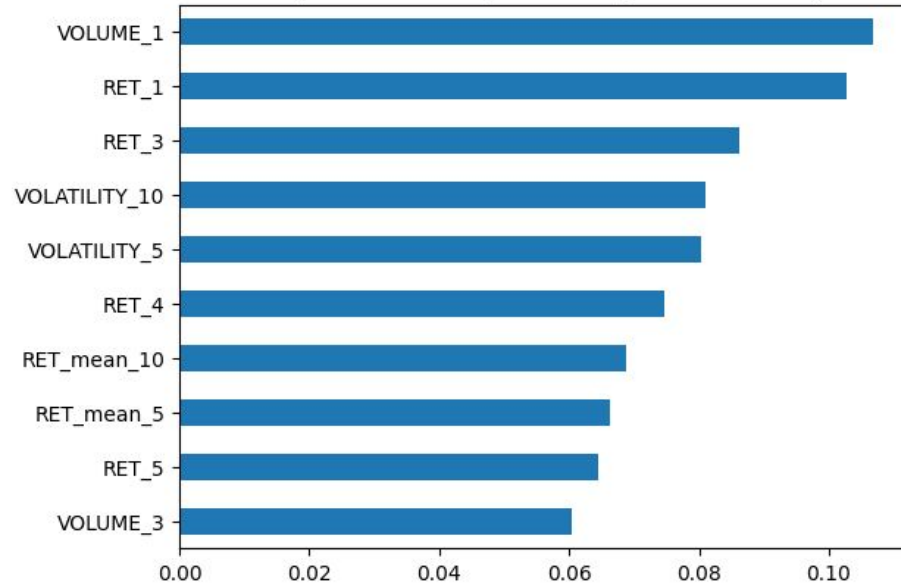
Our goal is to beat that score



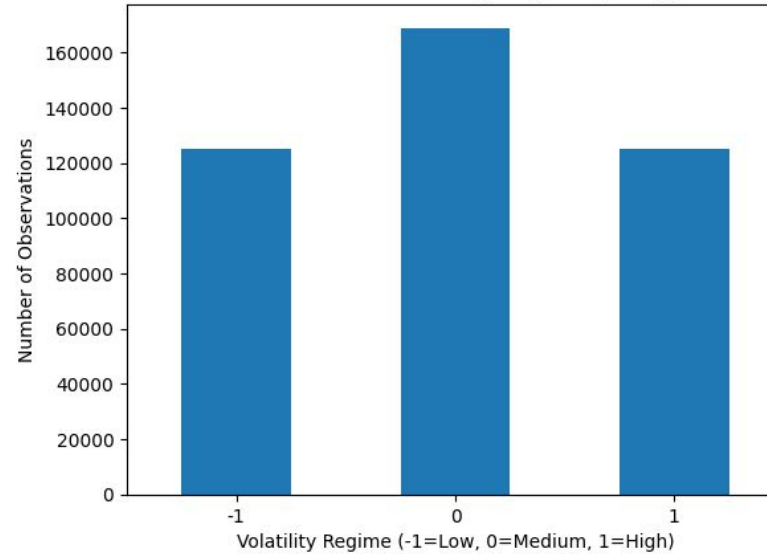
III- Features engineering

❖ Preparing Numerical Features for training

Top 10 Feature Importances (Random Forest)



Distribution of Volatility Regimes (5 day)



Raw numeric feature



Momentum (Short and long term)

Volatility

Regime of volatility



Improvement in the accuracy :

- RF (without/with FE) → 50,5% to 52,45%

Final numeric dataset: 15 meaningful features ready to be used

IV- Modeling Strategy & Ensemble Learning

A robust 5-fold cross-validation protocol is used to validate model stability and prevent overfitting on noise-heavy financial time series.

❖ 1. Multi-Model Approach:

Algorithm	Role & Justification	Regularization Strategy (Hyperparameters)
XGBoost <i>(Gradient Boosting)</i>	Role: Bias Reduction Sequentially corrects errors. Its built-in L1/L2 regularization is crucial for preventing overfitting in financial time series.	Slow Learning Strategy: • <code>learning_rate=0.03</code> • <code>subsample=0.7</code> <i>Logic:</i> A low learning rate combined with stochastic subsampling forces the model to generalize rather than memorize training data.
LightGBM <i>(Gradient Boosting)</i>	Role: Efficiency & Diversity Uses leaf-wise growth to capture different patterns than XGBoost. Highly optimized for speed and large datasets.	Convergence Strategy: • <code>n_estimators=800</code> • <code>colsample_bytree=0.7</code> <i>Logic:</i> We increased the number of trees to ensure effective convergence despite the aggressive regularization and slow learning rate.

IV- Modeling Strategy & Ensemble Learning

Algorithm	Role & Justification	Regularization Strategy (Hyperparameters)
Random Forest <i>(Bagging)</i>	Role: Variance Reduction Robust to noise and outliers by averaging multiple deep trees. Captures non-linear interactions without sensitive preprocessing.	Pruning Strategy: • <code>max_depth=10</code> • <code>min_samples_leaf=50</code> <i>Logic:</i> We strictly limited tree depth to force the model to ignore idiosyncratic noise and focus on stable, robust patterns.

❖ 2. Ensemble Strategy (Soft Voting)

Instead of relying on a single model, we leverage the "Wisdom of Crowds":

- **Technique:** Weighted Soft Voting (Average of predicted probabilities).

$$P_{final} = \frac{1}{3}(P_{RF} + P_{XGB} + P_{LGBM})$$

- **Decision Logic:**

- If $P_{final} > 0.5$ **→ Buy (True)**
- Else **→ Sell (False)**

- **Key Benefit:** Financial signals are weak; individual models make uncorrelated errors. Averaging them cancels out noise, resulting in a more robust generalization.

V - Results and comparison

❖ Our ranking

- **Baseline with LR → 50,5%**
- **MLP → 50,9%**
- **RF with FE → 52,45% (already 16th on the board)**
- **Final model → 58,31% (2nd on the board)**

Rang	Date	Participant(s)	Score public
1	13 février 2025 01:37	wojdalski	0,6306
2	14 décembre 2020 22:15	realFrancoisPignon2	0,5287
3	14 octobre 2024 15:01	duter	0,5265
4	14 octobre 2024 21:06	marco_hkm2	0,5265
5	27 avril 2025 15:37	Chenxin_nie	0,5263
6	11 décembre 2025 20:07	giantomm	0,5263
7	14 octobre 2024 15:40	Marco_hkm	0,5262
8	9 décembre 2024 23:35	Kyoukai	0,5256
9	10 mars 2025 14:08	wayancrain21	0,5254
10	28 mars 2022 12:46	lalohey	0,5254
11	26 avril 2025 17:25	papakon	0,5254
12	8 mars 2025 14:45	JuGir	0,5253
13	22 mai 2020 14:03	realFrancoisPignon	0,5252
14	9 juin 2025 10:54	ryanbty	0,5249
15	26 décembre 2024 18:23	nicolas_seinturier	0,5248
16	1 novembre 2025 12:38	aaiguepe	0,5245
17	19 mars 2025 21:56	wayancrain21_2	0,5244
18	13 octobre 2024 15:39	thomasvernet	0,5243
19	25 mars 2022 06:50	Prevision.io	0,5242
20	8 mars 2025 17:24	erraya	0,5242

VI - Challenging points and hindsight

❖ Acquired knowledge

- **The Noise Barrier:**
Markets are almost efficient. Predictive signals are often drowned out by random noise.
- **The Risk of Overfitting:**
A complex model can reach a good accuracy in training, but drop back in testing.
- **Favor robustness (simple features, ranking) over pure complexity.**



Thank you for your attention !